Logspace-uniform logarithmic-depth approximation of substochastic matrix powers and BPL inclusion

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doi:10.4230/LIPIcs.CCC.2024.32; CC BY4.0

Allowed standard prerequisites (editorial).

- Standard induced L1 matrix norm inequalities and telescoping difference-of-powers bound. Author Theorem5, exact statements retained.
- Standard probabilistic-space machine transition-matrix representation. Author introduction p.32:2, exact paragraph retained; author Corollary21 is the immediate stated consequence.
- Ajtai constant-depth approximate majority and convolution pairwise hashing. Author Lemma15 cites[2], Definition12/Theorem13 cites[22]; statements and exact construction retained with complete local approximate-counting proof.

Author circuit and hashing definitions

Definition 8 (AC circuit). AC circuit is a circuit with input gates, NOT gates, unbounded fan-in AND/OR gates, and (possibly more than one) output gates. The size of a circuit is defined by the number of AND/OR gates. The depth of a circuit is defined by the largest number of AND/OR gates on any path from an input gate to an output gate.

Definition 9 (L-uniform AC circuit family). For functions $S, d : \mathbb{Z}^+ \to \mathbb{R}^+$, we say a collection of circuits $\{C_n\}_{n\in\mathbb{Z}^+}$ is an S-size d-depth L-uniform AC circuit family, if each C_n has size $\leq S(n)$ and depth $\leq d(n)$, and given the binary representation of n, the description of C_n can be computed in uniform $O(\log n)$-space. We need to mention that the number of input gates in C_n is not necessarily n. Also note that since we can encode a tuple of $O(1)$ many integers to a single integer, we can also consider circuit collections with a tuple of integers as an index.

Definition 10 (Complexity Class L-AC^1). We say a language L is in the class L-AC^1 if there exists a poly(n)-size $O(\log n)$-depth L-uniform AC circuit family $\{C_n\}$ such that C_n computes L on n-bit inputs.

Definition 11 (GapMaj). For $n \in \mathbb{Z}^+$ and $a, b \in \mathbb{R}$ such that $0 \leq a < b \leq n$, define the promise problem GapMaj$[a, b]$ on n bits as follow:

$$\text{GapMaj}[a,b](x_1,\ldots,x_n) := \begin{cases} \text{YES} & \text{if } x_1,\ldots,x_n \text{ contains } \geq b \text{ 1's} \\ \text{NO} & \text{if } x_1,\ldots,x_n \text{ contains } \leq a \text{ 1's} \\ \perp & \text{otherwise.} \end{cases}$$

We will use pairwise independent hash function as a tool for approximate counting in AC. We shall use the following construction based on convolution, which was also used in [22].

Definition 12 (Convolution-Based Pairwise Independent Hash Function). Suppose m is a power of 2. Define $H_m : [m^3] \times [m] \to [m]$ by: for $(k, x) \in [m^3] \times [m]$, let $x_1 \cdots x_{\log m}$ be the binary representation of $x - 1$, let $a_1 \cdots a_{2\log m} b_1 \cdots b_{\log m}$ be the binary representation of $k - 1$, let $y_j := (\sum_{i=1}^{\log m} a_{i+j} x_i + b_j) \bmod 2$ for $j \in [\log m]$, then define $H_m(k, x)$ by letting $y_1 \cdots y_{\log m}$ be the binary representation of $H_m(k, x) - 1$.

Theorem 13. H_m is Pairwise Independent Hash Function in the following sense: for any $1 \leq i < j \leq m$, when k is sampled from the uniform distribution over $[m^3]$, the joint distribution of $(H_m(k, i), H_m(k, j))$ is identical to the uniform distribution over $[m] \times [m]$.

Author matrix conventions and complexity context

One may view derandomizing BPL as the problem of approximating powers of substochastic matrices. For a TM with s bits of memory, one can label all its states by elements in $[2^s]$. We can define $A \in \mathbb{R}^{2^s \times 2^s}$ to be its transition matrix in the sense that $A_{i,j}$ is the probability that the machine moves from state i to state j in one step. Note that it must arrive at an accept or reject state in 2^s steps, so we only need to approximate A^{2^s}.

Definition 4 (L_1-norm). Define the L_1-norm of a vector $(x_1, \ldots, x_n)^\top \in \mathbb{R}^n$ to be

$$\|(x_1,\ldots,x_n)^\top\|_1 := |x_1| + \cdots + |x_n|.$$

Define the L_1-norm of a matrix $A \in \mathbb{R}^{n \times n}$ to be

$$\|A\|_1 := \sup_{x \in \mathbb{R}^n} \frac{\|Ax\|_1}{\|x\|_1} = \max_{1 \leq j \leq n} \{|A_{1,j}| + |A_{2,j}| + \cdots + |A_{n,j}|\}.$$

Theorem 5. For any $A, B \in \mathbb{R}^{n \times n}$, we have:

1. $\|A + B\|_1 \leq \|A\|_1 + \|B\|_1$;
2. $\|AB\|_1 \leq \|A\|_1 \|B\|_1$;
3. If $\|A\|_1, \|B\|_1 \leq 1$, then for any $p \in \mathbb{Z}^+$, $\|A^p - B^p\|_1 \leq p\|A - B\|_1$.

Definition 6 (Non-negative Matrix). We say a matrix is non-negative if each of its entry is non-negative.

Definition 7 (Substochastic Matrix). We say a matrix $A \in \mathbb{R}^{n\times n}$ is a substochastic matrix if A is non-negative and $\|A\|_1 \leq 1$. For simplicity, we always assume that the size of a substochastic matrix is a power of 2. To represent a substochastic matrix, we independently represent each entry in binary, accurate to $100 \log n$ decimal places.

Theorem 20. Let n be a power of 2. Then there exists a $\mathrm{poly}(n)$-size $O(\log n)$-depth L-uniform AC circuit family $\{\mathcal{M}_n\}$[1] such that on input a substochastic matrix $A \in \mathbb{R}^{n\times n}$, $\mathcal{M}_n$ outputs a substochastic matrix $M \in \mathbb{R}^{n\times n}$ such that $\|M - A^n\|_1 \leq 1/n$.

Corollary 21. $\mathrm{BPL} \subseteq \mathrm{L\text{-}AC}^1$.

Theorem 14. Let $n, a, b \in \mathbb{N}$ such that $0 \leq a < b \leq n$. Then there exists a $\mathrm{poly}(n)$-size $O\left(\frac{\log \frac{b}{b-a}}{\log\log n} + 1\right)$-depth L-uniform AC circuit family $\{C_{n,a,b}\}$ that computes $\mathrm{GapMaj}[a, b]$ on n bits.

The proof depends on the next few Lemmas.

Lemma 15 ([2]). Let $n \in \mathbb{Z}^+$. Then there exists $\mathrm{poly}(n)$-size $O(1)$-depth L-uniform AC circuit family $\{C_n^{(0)}\}$ that computes $\mathrm{GapMaj}[n/3, 2n/3]$ on n bits.

Lemma 16 (Exact Counting). Let $n, \ell \in \mathbb{Z}^+$ such that $n \geq \ell$. Then there exists a $\mathrm{poly}(n)$-size $O\left(\frac{\log \ell}{\log\log n} + 1\right)$-depth L-uniform AC circuit family $\{E_{n,\ell}\}$ such that on ℓ bits of input, $E_{n,\ell}$ outputs the exact number of 1's over the input bits, in binary form. We remark that in Lemma 16, n is only used to bound the size of the circuit. Also, n, ℓ are not necessarily polynomially related.

Proof. We only need to show how to compute sum of $O(\sqrt{\log n})$ many $O(\log n)$-bit[2] non-negative integers in $O(1)$-depth, then by divide-and-conquer we can compute sum of ℓ bits in $O\left(\frac{\log \ell}{\log\log n} + 1\right)$-depth. View the $O(\log n)$-bit integers as $2^{\lfloor\sqrt{\log n}\rfloor}$-base $O(\sqrt{\log n})$-digit integers. We use the grade-school algorithm to sum $O(\sqrt{\log n})$ integers as follow. We first guess the result and all carry-bits, which involve at most $O(\sqrt{\log n}) \cdot O\left(\log\left(\sqrt{\log n} \cdot 2^{\lfloor\sqrt{\log n}\rfloor}\right)\right) = O(\log n)$ bits, and thus has at most $\mathrm{poly}(n)$ choices. Then we can apply a local check on each digit, each local check involves at most $O(\log n)$ bits, and thus deciding whether all local checks are passed can be computed in $O(1)$-depth. Then we can take the result of the only guess that passes all local checks. The total cost is $O(1)$-depth. □

[1] We require that given n, description of $\mathcal{M}_n$ can be computed in space $O(\log n)$.

[2] Here "$O(\log n)$-bit integers" refers to integers which has $O(\log n)$-bits in its binary representation.

Lemma 17. Let $n, a, b \in \mathbb{N}$ such that $0 \le a < b \le n$. Then there exists a poly(n)-size $O\left(\frac{\log \frac{n}{b-a}}{\log\log n} + 1\right)$-depth L-uniform AC circuit family $\{C^{(1)}_{n,a,b}\}$ that computes GapMaj$[a, b]$ on n bits.

Proof. Only consider the case that n is a power of 2, otherwise we can use a simple padding argument. By Lemma 15, it suffices to show how to reduce GapMaj$[a, b]$ on n bits to GapMaj$[n^3/3, 2n^3/3]$ on n^3 bits, via a poly(n)-size $O\left(\frac{\log \frac{n}{b-a}}{\log\log n} + 1\right)$-depth L-uniform AC circuit. If $b - a \le 4\sqrt{n}$ then we can directly compute the number of 1's exactly via Lemma 16. Below we only consider $b - a > 4\sqrt{n}$. Let $\ell := \left\lceil \frac{12n^2}{(b-a)^2} \right\rceil$. Suppose the GapMaj$[a, b]$ instance is $x_1, x_2, \ldots, x_n$. Let H_n be the hash function defined in Definition 12. Define $y_1, \ldots, y_{n^3}$ as follow: for $i \in [n^3]$, let y_i be 1 if at least $\frac{a+b}{2n}$ fraction of $x_{H_n(i,1)}, \ldots, x_{H_n(i,\ell)}$ is 1, otherwise let y_i be 0. Note that $y_1, \ldots, y_{n^3}$ can be computed via a poly(n)-size $O\left(\frac{\log \ell}{\log\log n} + 1\right)$-depth L-uniform AC circuit, by Lemma 16. Here $O\left(\frac{\log \ell}{\log\log n} + 1\right) = O\left(\frac{\log \frac{n}{b-a}}{\log\log n} + 1\right)$. Let's do some simple calculations. Assume p fraction of $x_1, \ldots, x_n$ is 1. Let S_i be number of 1's in $x_{H_n(i,1)}, \ldots, x_{H_n(i,\ell)}$. Then we have $\mathbb{E}_{i\sim[n^3]}[S_i] = p\ell$ and $\mathrm{Var}_{i\sim[n^3]}[S_i] \le \ell$. So if $p \le \frac{a}{n}$, then

$$\Pr_{i\sim[n^3]}\left[S_i \ge \ell \cdot \frac{a+b}{2n}\right] \le \frac{\ell}{(\ell \cdot \frac{b-a}{2n})^2} = \frac{4n^2}{\ell(b-a)^2} \le \frac{1}{3}.$$

Similarly if $p \ge \frac{b}{n}$ then

$$\Pr_{i\sim[n^3]}\left[S_i \le \ell \cdot \frac{a+b}{2n}\right] \le \frac{1}{3}.$$

This means if $x_1, \ldots, x_n$ is YES/NO instance of GapMaj$[a, b]$, then $y_1, \ldots, y_{n^3}$ is YES/NO instance of GapMaj$[n^3/3, 2n^3/3]$. The reduction is completed. □

Proof of Theorem 14. We will try to reduce to Lemma 17. Suppose the GapMaj$[a, b]$ instance is $x_1, x_2, \ldots, x_n$. We only consider the case n is a power of 2, otherwise use a simple padding argument. We only consider the case $10\left(\frac{b}{b-a}\right)^2 < \frac{n}{b-a}$ (or equivalently, $n(b-a) > 10b^2$), otherwise we can directly apply Lemma 17. Let $\ell := \left\lfloor \frac{n(b-a)}{2b^2} \right\rfloor$. For $i \in [n^3]$, let $y_i := x_{H_n(i,1)} \vee \cdots \vee x_{H_n(i,\ell)}$, here H_n is the hash function defined in Definition 12. Then $y_1, \ldots, y_{n^3}$ can be computed via poly(n)-size $O(1)$-depth L-uniform AC circuit. Assume p fraction of $x_1, \ldots, x_n$ is 1. Let S_i be number of 1's in $x_{H_n(i,1)}, \ldots, x_{H_n(i,\ell)}$. Then we have $\mathbb{E}_{i\sim[n^3]}[S_i] = p\ell$ and $\mathbb{E}_{i\sim[n^3]}[S_i^2] = \ell(\ell-1)p^2 + \ell p \le \ell p + \ell^2 p^2$. Thus by

$$\frac{\mathbb{E}_{i\sim[n^3]}[S_i]^2}{\mathbb{E}_{i\sim[n^3]}[S_i^2]} \le \Pr_{i\sim[n^3]}[S_i \ge 1] \le \mathbb{E}_{i\sim[n^3]}[S_i]$$

we know: if $p \le \frac{a}{n}$, then $\Pr_{i\sim[n^3]}[S_i \ge 1] \le \frac{\ell a}{n}$; if $p \ge \frac{b}{n}$, then

$$\Pr_{i\sim[n^3]}[S_i \ge 1] \ge \frac{(\frac{\ell b}{n})^2}{\frac{\ell b}{n} + (\frac{\ell b}{n})^2} \ge \frac{\ell b}{n} - \left(\frac{\ell b}{n}\right)^2.$$

To summarize, if $x_1, \ldots, x_n$ is YES/NO instance of GapMaj$[a, b]$, then $y_1, \ldots, y_{n^3}$ is YES/NO instance of

$$\text{GapMaj}\left[\left\lfloor n^3 \cdot \frac{\ell a}{n}\right\rfloor, \left\lceil n^3 \cdot \left(\frac{\ell b}{n} - \left(\frac{\ell b}{n}\right)^2\right)\right\rceil\right].$$

Finally we observe that

$$\left(\frac{\ell b}{n} - \left(\frac{\ell b}{n}\right)^2\right) - \frac{\ell a}{n} = \ell \cdot \left(\frac{b-a}{n} - \frac{\ell b^2}{n^2}\right) \geq \frac{n(b-a)}{3b^2} \cdot \frac{b-a}{2n} = \frac{(b-a)^2}{6b^2}.$$

Thus by Lemma 17, GapMaj $\left[\left\lfloor n^3 \cdot \frac{\ell a}{n}\right\rfloor, \left\lceil n^3 \cdot \left(\frac{\ell b}{n} - \left(\frac{\ell b}{n}\right)^2\right)\right\rceil\right]$ over n^3 bits can be computed via a poly(n)-size $O\left(\frac{\log \frac{b}{b-a}}{\log\log n} + 1\right)$-depth L-uniform AC circuit. $\square$

4 The Iteration Method

In this section, we will introduce the iteration step, which is the core of our proof of BPL $\subseteq$ L-AC^1.

Theorem 18 (The Iteration). Let $A \in \mathbb{R}^{n\times n}$ be a substochastic matrix and $k, t \in \mathbb{Z}^+$ such that $\log n \geq k \geq t$. Suppose substochastic matrices $B_0, \ldots, B_{k-1}$ are approximations of $A^{2^0}, \ldots, A^{2^{k-1}}$ such that $\|B_i - A^{2^i}\|_1 \leq \varepsilon_i$ for $i = 1, 2, \ldots, k-1$. Define

$$\begin{aligned} C := &- \sum_{i=1}^{t-1} \sum_{\substack{\{j_1<\cdots<j_p\}\uplus\{j'_1<\cdots<j'_q\}\\ =\{k-1,k-2,\ldots,k-i+1\}}} B_{j_p}\cdots B_{j_1} B_{k-i}^2 B_{j'_1}\cdots B_{j'_q} \\ &+ \sum_{\substack{\{j_1<\cdots<j_p\}\uplus\{j'_1<\cdots<j'_q\}\\ =\{k-1,k-2,\ldots,k-t+1\}}} B_{j_p}\cdots B_{j_1} B_{k-t}^2 B_{j'_1}\cdots B_{j'_q}. \end{aligned}$$

Then

$$\|C - A^{2^k}\|_1 \leq \sum_{i=1}^{t-1} 2^{i-1}\varepsilon_{k-i}^2 + 2^t \varepsilon_{k-t}.$$

Proof. Note that

$$\begin{aligned} C - A^{2^k} = &- \sum_{i=1}^{t-1} \sum_{\substack{\{j_1<\cdots<j_p\}\uplus\{j'_1<\cdots<j'_q\}\\ =\{k-1,k-2,\ldots,k-i+1\}}} B_{j_p}\cdots B_{j_1} (A^{2^{k-i}} - B_{k-i})^2 B_{j'_1}\cdots B_{j'_q} \\ &- \sum_{\substack{\{j_1<\cdots<j_p\}\uplus\{j'_1<\cdots<j'_q\}\\ =\{k-1,k-2,\ldots,k-t+1\}}} B_{j_p}\cdots B_{j_1} (A^{2^{k-t+1}} - B_{k-t}^2) B_{j'_1}\cdots B_{j'_q}. \end{aligned}$$

So by Theorem 5,

$$\begin{aligned}\|C - A^{2^k}\|_1 &\le \sum_{i=1}^{t-1} 2^{i-1}\|A^{2^{k-i}} - B_{k-i}\|_1^2 + 2^t\|A^{2^{k-t}} - B_{k-t}\|_1 \\ &\le \sum_{i=1}^{t-1} 2^{i-1}\varepsilon_{k-i}^2 + 2^t\varepsilon_{k-t}.\end{aligned}$$

□

Theorem 19 (Computing the Iteration). Let $n, k, t, A, B_0, \ldots, B_{k-1}, \varepsilon_0, \ldots, \varepsilon_{k-1}, C$ be as defined in Theorem 18. Let d be an integer such that $4\log n \ge d \ge t/10$. Then there exists a poly(n)-size $O(d)$-depth L-uniform AC circuit family $\{\mathcal{I}_{n,k,t,d}\}$ such that on inputs $B_{k-t}, \ldots, B_{k-1}$, if

$$\sum_{i=1}^{t-1} 2^{i-1}\varepsilon_{k-i}^2 + 2^t\varepsilon_{k-t} \le \frac{1}{2^{d+2}}$$

is satisfied, then $\mathcal{I}_{n,k,t,d}$ outputs a substochastic matrix C' such that $\|C' - A^{2^k}\|_1 \le 1/2^d$. The intuition behind Theorem 19 is that to approximately compute C, all arithmetic operations only need a multiplicative accuracy of $1/2^{\Theta(d)}$. This can be done efficiently by L-uniform AC circuit by Theorem 14.

Proof of Theorem 19. We observe that C is the sum of 2^{t-1} "+" terms and $2^{t-1} - 1$ "-" terms, and each term is a multiplication of not more than $t + 1$ substochastic matrices. We will first show how to approximate the multiplication of substochastic matrices and then show how to approximate their sum. To approximate $Z := XY$ for two substochastic matrices X, Y, we only need to approximate $\sum_{r=1}^n X_{i,r}Y_{r,j}$ for each pair $(i, j) \in [n]^2$. We first represent each entry $X_{i,r}, Y_{r,j}$ using n^{100} bits such that fraction of 1's in these n^{100} bits is equal to the entry, then use a layer of AND gate to represent each $X_{i,r}Y_{r,j}$ using fraction of 1's in n^{200} bits, and then represent each $\frac{1}{n}\sum_{r=1}^n X_{i,r}Y_{r,j}$ using fraction of 1's in n^{201} bits. Then we invoke $C_{n^{201},\ell,\lceil\ell(1+1/2^{20d+10})\rceil}$ (as defined in Theorem 14, which has depth $\le O(\frac{d}{\log\log n} + 1) \le O(\frac{d}{\log(t+1)})$)[3] for $\ell = 1, 2, \ldots, n^{200}$ over these n^{201} bits. Suppose ℓ_0 is the smallest index such that $C_{n^{201},\ell_0,\lceil\ell_0(1+1/2^{20d+10})\rceil}$ outputs 0, then we have

$$\frac{\ell_0 - 1}{n^{200}} < Z_{i,j} < \frac{\ell_0(1 + \frac{1}{2^{20d+10}})}{n^{200}}$$

and thus[4]

$$\frac{Z_{i,j}}{1 + \frac{1}{2^{20d+10}}} - \frac{1}{n^{100}} \le \frac{1}{n^{100}}\left[\frac{\ell_0}{n^{100}}\right] \le Z_{i,j}.$$

Use $[\ell_0/n^{100}]/n^{100}$ as an approximation of $Z_{i,j}$, then we obtain an approximation $\widetilde{Z}$ of Z such that $Z - \widetilde{Z}$ is non-negative and $\widetilde{Z}$ is substochastic and $\|Z - \widetilde{Z}\|_1 \le 1/2^{20d+10} + 1/n^{99}$. We need to be careful that here we need

[3] In Theorem 14 we take $(a, b) = (\ell, \lceil\ell(1 + 1/2^{20d+10})\rceil)$, and then $\log\frac{b}{b-a} \le O(d)$.

[4] Since $n^{200}Z_{i,j}$ is an integer, we have $\frac{\ell_0-1}{n^{200}} < Z_{i,j} \Longrightarrow \frac{\ell_0}{n^{200}} \le Z_{i,j}$.

a multiplicative small error on each entry and thus we need to strengthen Lemma 17 to Theorem 14. Then multiplication of not more than $t+1$ substochastic matrices can be computed via $O(\log(t+1))$ layers of multiplication of two matrices. Recall that multiplying two matrices uses $O(\frac{d}{\log(t+1)})$-depth and has additive error $1/2^{20d+10} + 1/n^{99}$. So the total depth for computing multiplication of not more than $t+1$ substochastic matrices is $O(d)$ and the total error is $\leq t(1/2^{20d+10} + 1/n^{99}) \leq 1/2^{19d+5}$. To summarize, suppose $C = -\sum_{i=1}^{2^{t-1}-1} D_i + \sum_{i=1}^{2^{t-1}} D'_i$, here each D_i, D'_i is multiplication of some substochastic matrices. Then we can compute their approximations $\widetilde{D}_i, \widetilde{D}'_i$ in $O(d)$ depth such that $\|D_i - \widetilde{D}_i\|_1 \leq 1/2^{19d+5}$ and $\|D'_i - \widetilde{D}'_i\|_1 \leq 1/2^{19d+5}$. We approximate $\frac{1}{2^{t-1}}\sum_{i=1}^{2^{t-1}-1} \widetilde{D}_i$ and $\frac{1}{2^{t-1}}\sum_{i=1}^{2^{t-1}} \widetilde{D}'_i$. Use the similar idea as summing $\frac{1}{n}\sum_{r=1}^{n} X_{i,r}Y_{r,j}$, we can compute substochastic matrices C^-, C^+ using $O(d)$-depth, such that

$$\left\| C^- - \frac{1}{2^{t-1}} \sum_{i=1}^{2^{t-1}-1} \widetilde{D}_i \right\|_1 \leq \frac{1}{2^{19d+5}},$$
$$\left\| C^+ - \frac{1}{2^{t-1}} \sum_{i=1}^{2^{t-1}} \widetilde{D}'_i \right\|_1 \leq \frac{1}{2^{19d+5}}.$$

Then $2^{t-1}(C^+ - C^-)$ is a good approximation of A^{2^k} since

$$\begin{aligned}
\|2^{t-1}(C^+ - C^-) - A^{2^k}\|_1 \leq{}& 2^{t-1}\left\| C^- - \frac{1}{2^{t-1}} \sum_{i=1}^{2^{t-1}-1} \widetilde{D}_i \right\|_1 + 2^{t-1}\left\| C^+ - \frac{1}{2^{t-1}} \sum_{i=1}^{2^{t-1}} \widetilde{D}'_i \right\|_1 \\
&+ \sum_{i=1}^{2^{t-1}-1} \|D_i - \widetilde{D}_i\|_1 + \sum_{i=1}^{2^{t-1}} \|D'_i - \widetilde{D}'_i\|_1 \\
&+ \left\| -\sum_{i=1}^{2^{t-1}-1} D_i + \sum_{i=1}^{2^{t-1}} D'_i - A^{2^k} \right\|_1 \\
\leq{}& \frac{2^{t-1}}{2^{19d+5}} + \frac{2^{t-1}}{2^{19d+5}} + \frac{2^{t-1}}{2^{19d+5}} + \frac{2^{t-1}}{2^{19d+5}} + \|C - A^{2^k}\|_1 \\
\leq{}& \frac{1}{2^{9d+4}} + \left(\sum_{i=1}^{t-1} 2^{i-1}\varepsilon_{k-i}^2 + 2^t \varepsilon_{k-t} \right) \\
\leq{}& \frac{1}{2^{9d+4}} + \frac{1}{2^{d+2}}.
\end{aligned}$$

Here the last step is from the statement of Theorem 19. Finally we compute a substochastic matrix C' which is a good approximation of A^{2^k} and $2^{t-1}(C^+ - C^-)$. Here we need to be careful that C and $2^{t-1}(C^+ - C^-)$ are not necessarily non-negative or substochastic (but A^{2^k} is guaranteed substochastic). Let

$$\begin{aligned}
C''_{i,j} &:= \max\{2^{t-1}(C^+_{i,j} - C^-_{i,j}), 0\}, \\
C'_{i,j} &:= \frac{1}{n^{100}} \left[C''_{i,j} \left(1 - \frac{1}{2^{d+1}} \right) \cdot n^{100} \right].
\end{aligned}$$

We can compute C' given C^+, C^- by hardwiring the map $(C^+_{i,j}, C^-_{i,j}) \mapsto C'_{i,j}$, which is L-uniform. Obviously C' is non-negative. Note that C'' is entrywise closer to A^{2^k} than $2^{t-1}(C^+ - C^-)$ and hence

$$\|C'' - A^{2^k}\|_1 \le \|2^{t-1}(C^+ - C^-) - A^{2^k}\|_1 \le \frac{1}{2^{9d+4}} + \frac{1}{2^{d+2}}.$$

Therefore C' is substochastic since

$$\|C'\|_1 \le \left(1 - \frac{1}{2^{d+1}}\right)\|C''\|_1 \le \left(1 - \frac{1}{2^{d+1}}\right)\left(1 + \frac{1}{2^{9d+4}} + \frac{1}{2^{d+2}}\right) \le 1.$$

Also note that

$$\begin{aligned}
\|C' - A^{2^k}\|_1 &\le \|C' - C''\|_1 + \|C'' - A^{2^k}\|_1 \\
&\le \frac{1}{n^{99}} + \frac{1}{2^{d+1}}\|C''\|_1 + \|C'' - A^{2^k}\|_1 \\
&\le \frac{1}{n^{99}} + \frac{1}{2^{d+1}}\left(1 + \frac{1}{2^{9d+4}} + \frac{1}{2^{d+2}}\right) + \frac{1}{2^{9d+4}} + \frac{1}{2^{d+2}} \\
&\le \frac{1}{2^d}.
\end{aligned}$$

To summarize, we can output a valid C' in $O(d)$-depth. And the circuit is poly(n)-size and L-uniform. □

5 The Complete Algorithm

Proof of Theorem 20. Only consider $\log n \ge 10$. For $k, t \in \mathbb{N}$ such that $k \le \log n$ and $1 \le t \le 3\log n - 2k$, we wish to compute a substochastic matrix $M(k,t)$, which is an approximation of A^{2^k}, such that $\|M(k,t) - A^{2^k}\|_1 \le 1/2^t$. Then $M := M(\log n, \log n)$ is the desired matrix. For $k = 0$, we can trivially let $M(0,t) := A$. Now we show how to recursively compute $M(k_0, t_0)$ for $k_0 = 1, 2, \ldots, \log n$. In Theorem 18, take the same n, A and take $k := k_0$, take $B_{k-i} := M(k-i, [t_0/2]+2i)$ for $1 \le i \le k$. Then we can take $\varepsilon_{k-i} := 1/2^{[t_0/2]+2i}$ for $1 \le i \le k-1$ and $\varepsilon_0 = 0$. Now we will invoke Theorem 18,19 by choosing parameter t properly according to the following two cases.

Case 1. $k \le 2t_0 + 2$. Take the parameter t in Theorem 18 to be $t := k$. Then

$$\sum_{i=1}^{k-1} 2^{i-1}\varepsilon^2_{k-i} + 2^k\varepsilon_0 = \sum_{i=1}^{k-1} \frac{1}{2^{2[t_0/2]+3i+1}} \le \frac{1}{2^{t_0+2}}.$$

In Theorem 19 take $d := t_0$. It is easy to verify that $\log n \ge k \ge t$ and $4\log n \ge d \ge t/10$ hold when we invoke Theorem 18,19. Given $B_{k-1}, \ldots, B_0$, use $\mathcal{I}_{n,k_0,k_0,t_0}$ (defined in Theorem 19) we can compute a substochastic matrix C' such that $\|C' - A^{2^k}\|_1 \le 1/2^{t_0}$.

Case 2. $k \ge 2t_0 + 3$. Take $t := 2t_0 + 3$ in Theorem 18. Then

$$\sum_{i=1}^{2t_0+2} 2^{i-1}\varepsilon^2_{k-i} + 2^{2t_0+3}\varepsilon_{k-2t_0-3} \le \sum_{i=1}^{2t_0+2} \frac{1}{2^{2[t_0/2]+3i+1}} + \frac{1}{2^{[t_0/2]+2t_0+3}} \le \frac{1}{2^{t_0+2}}.$$

In Theorem 19 take $d := t_0$. Given $B_{k-1}, \ldots, B_0$, use $\mathcal{I}_{n,k_0,2t_0+3,t_0}$ we can compute a substochastic matrix C' such that $\|C' - A^{2^k}\|_1 \leq 1/2^{t_0}$. To summarize, take $M(k_0, t_0) := C'$, we can compute $M(k_0, t_0)$ given $M(k_0 - i, [t_0/2] + 2i)$ for $1 \leq i \leq k_0$, via a poly(n)-size $O(t_0)$-depth L-uniform AC circuit. Let $\gamma > 0$ be a concrete constant such that we can compute $M(k_0, t_0)$ given $M(k_0 - i, [t_0/2] + 2i)$'s via a poly(n)-size γt_0-depth L-uniform AC circuit. Note that if $M(k_0 - i, [t_0/2] + 2i)$ can be computed in $2\gamma(2(k_0 - i) + ([t_0/2] + 2i))$-depth for $1 \leq i \leq k_0$, then $M(k_0, t_0)$ can be computed in

$$\gamma t_0 + \max_{1 \leq i \leq k_0} \{2\gamma(2(k_0 - i) + ([t_0/2] + 2i))\} \leq 2\gamma(2k_0 + t_0)$$

-depth. Also note that $M(0, t_0)$'s are just the inputs, so by induction we know $M(k_0, t_0)$ can be computed in $2\gamma(2k_0 + t_0)$-depth. Specially, $M(\log n, \log n)$ (which is the desired output) can be computed in $6\gamma \log n \leq O(\log n)$-depth. Also note that we use "compute $M(k_0, t_0)$ given $M(k_0 - i, [t_0/2] + 2i)$" $O((\log n)^2)$ many times, so the total circuit size for computing $M(\log n, \log n)$ is still poly(n). □

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